FFNOML

Julia 3D Inversion Implementation Report

As-built architecture, verification evidence, engineering decisions, known limits, and the Phase 6 handoff state. Companion to development plan v1.4.

PURPOSE

The planning PDF describes intended work. This report describes the code that now exists, the exact local evidence obtained, implementation decisions that refine the plan, and tests that still require the target GPU or cluster. It is written as both a human review document and a machine-oriented continuation record for Codex.

Current conclusion

Phases 0 through 6 are implemented. Phase 6 adds the production matrix-free gradient chain, bounded L-BFGS, separate H/Ca population workspaces, rank-0 FFNO VJPs, HDF5 two-input/two-output execution, and mixed non-LTE plus Kurucz LTE gradient validation. Olivia job 2072078 accepted the solver, recovery, memory and CUDA/NCCL infrastructure; one new real-checkpoint FFNO VJP regression remains to be rerun.

Document	Implementation baseline	Prepared
Implementation report v1.2	Code and evidence through Phase 6	31 Aug 2026

1. Executive implementation status

The package root is **FFNOInversion.jl**. There is one forward architecture: MPI distributes 2D tiles, Julia threads execute independent local columns, and global rank 0 alone controls FFNO GPU work. A one-rank run is the smallest topology of the same path, not an alternate serial implementation.

Phase	State	Implemented outcome	Acceptance boundary
0 - Foundation	Accepted	Canonical arrays, nodes, configuration, observation weights, regularization, two-input/two-output contract, checkpoints, future capability seams.	Core and configuration tests pass.
1 - HE3D/MHS	Accepted	3D coupled pressure relaxation, Wittmann EOS, STiC 500 nm opacity, optional-B mode selection, Lorentz term, convergence diagnostics.	Manufactured and production-backend tests pass.
2 - FFNO	Accepted	Persistent PythonCall bridge, metadata/hash checks, recorded backend, real H and Ca checkpoint loading and parity.	CPU functional parity passes; target GPU performance pending.
3 - Synthesis	Accepted	Mixed FFNO non-LTE and Kurucz LTE opacity contributors with one formal solve; Muspel production adapter.	Halpha/Ca reference parity and Kurucz source-level tests pass.
4 - Hybrid path	Locally accepted	Distributed force balance, populations, threaded synthesis, PSF, chi-square, regularization, collection, failure recovery, provenance and timing.	Target GPU and multi-node scale evidence still external.
5 - Inversion	Accepted locally	Bounded/scaled coarse controls, distributed objective, pattern search, finite-difference oracle, diagnostics and portable restart.	Exact-model recovery and 1-rank/4-rank solver parity pass.
6 - Gradients/solver	Implemented	Production PSF, transfer, line, FFNO and force/regularization gradient chain; bounded L-BFGS; HDF5 executable path.	220 local assertions, mixed-gradient oracle parity, MPI VJP/restart and Olivia job 2072078 pass; updated real-checkpoint GPU VJP rerun pending.

What the local evidence proves

- The complete scientific result is topology invariant for the tested one-rank and four-rank layouts on an uneven 11 x 7 domain.
- Non-root ranks own only local atmospheric, population, synthesis, and halo arrays during the scientific hot path.
- The rank-0 GPU-control protocol propagates an injected failure to all ranks and the communicator remains usable for a subsequent launch.
- Checkpoints are independent of the MPI layout used to write or restore them.
- Repeated threaded evaluations retain workspace object identities and return identical spectra.
- Two distinct initial atmospheres recover the same temperature/velocity truth; interrupted and uninterrupted solver histories agree, including across MPI topology changes.

What it does not prove

Local tests do not establish scientific accuracy against an independent RH/MULTI3D forward model or the best production rank/thread topology. The updated real-checkpoint FFNO VJP must also be rerun once on Olivia after the Phase 6 implementation change. These boundaries are listed explicitly in Section 14.

2. As-built end-to-end architecture

The runtime follows the scientific dependencies without introducing a second execution layer. Data become rank-owned immediately after input distribution and remain so until explicit root collection for the present output adapter.

Rank 0 reads two inputs -> **MPI 2D tile distribution** -> **HE3D or MHS plus EOS** -> **Rank 0 FFNO control**

Population tile redistribution -> **Threaded local mixed synthesis** -> **Halo PSF plus global objective** -> **Rank 0 writes two outputs**

Runtime ownership

Owner	Long-lived state	Permitted transient state	Forbidden
Every MPI rank	Its non-empty x/y tile; local atmosphere, populations, spectra, observation, thread-private synthesis workspaces.	Halo buffers and small reduction values.	A complete CPU atmosphere after initial distribution.
Global rank 0	Same local tile plus GPU control endpoint and run metadata.	Complete atmosphere while staging an FFNO request; complete populations before scattering; complete final products for current output writing.	Launching a second independent scientific architecture.
Julia worker thread	One private synthesis workspace selected by thread id.	Column-local scalar/vector temporaries.	MPI calls or shared mutable scratch arrays.
GPU service	Persistent checkpoint, normalization, model and device buffers.	Full FFNO feature and population volumes when the selected backend requires global context.	Per-pixel inference that removes the learned 3D context.

Single production call

```
forward!(workspace, model, distributed_atmosphere, parallel_context) -> (spectrum, force_balance,
timings)
```

Node expansion, HE3D/MHS, population staging, synthesis, observation degradation, chi-square and regularization all consume distributed objects. MPI is therefore the architecture, including when communicator size is one.

3. Stable scientific and file contracts

The implementation preserves the project owner's operational model: two primary scientific inputs and two primary scientific outputs. Configuration selects datasets and algorithms; it is not a third data cube.

Contract	Contents	Current implementation note
Input 1: observation	Spectrum shaped (wavelength, Stokes, x, y), uncertainty, wavelength-by-Stokes weights, and 2D spatial weights.	Weights select chi-square participation. Zero means excluded; no interpolation or implicit downsampling.
Input 2: initial atmosphere	log tau500, temperature, vx/vy/vz, microturbulence, top gas pressure, and optional Bx/By/Bz.	The model grid can be finer than the observed active pixels; the weights alone choose fitted samples.
Output 1: synthesis	Full high-resolution PSF-degraded spectrum, wavelength coordinates, residual/chi-square diagnostics, provenance.	Current Phase 4 adapter collects on rank 0 before writing.
Output 2: atmosphere	Recovered atmosphere plus Pgas, rho, ne, z, populations and convergence/provenance.	Checkpoint/log files remain operational sidecars, not extra scientific outputs.

Simultaneous spectral regions

A region declares its wavelength grid and an ordered list of sources. FFNO transitions and Kurucz LTE lines may contribute to the same or different regions; extinction and emissivity are summed before one formal solution and the shared observation operator.

```
[[regions]]
start_angstrom = 8540.23102
step_angstrom = 0.04267
points = 88
continuum = 4.227725e-05
psf_type = "fpi"
psf_file = "8542.nc"

[[regions.sources]]
type = "ffno"
species = "Ca"
transition = "8542"

[[regions.sources]]
type = "kurucz_lte"
line_list = "kurucz_6301_6302.input"
```

Regularization

- Vertical slots retain the STiC-style order: temperature, vz, microturbulence, magnetic strength, inclination, azimuth, and gas-pressure boundary.
- Types are 0 none, 1 first derivative, 2 depth-mean deviation, 3 zero deviation, and 4 second derivative. Invalid temperature mean/zero requests normalize to none.
- Horizontal penalties are separate per variable and use MPI halos so derivatives cross tile boundaries.
- Normalization scales are distinct from penalty strengths; obsolete duplicate configuration weights were removed.

Future compatibility

SpectralCube always retains a Stokes axis and redistribution is an abstract policy. Release 1 activates intensity and non-PRD only. Unsupported PRD or Q/U/V requests fail capability validation; future implementations attach at existing seams rather than changing shapes or optimizer contracts.

4. Phase 0 - foundation implemented

Phase 0 converted the design discussion into executable types, schemas, mock physics, objectives, checkpoints, and architecture decisions.

Area	As-built decision	Evidence
Coordinates and units	Atmosphere (nz,nx,ny); spectrum (n_lambda,n_stokes,nx,ny); SI public boundaries; strictly monotonic log tau500.	Shape, monotonicity and type tests.
Node maps	Vertical node interpolation and tile-local expansion use global normalized x/y coordinates.	Node exactness and monolithic/tile parity tests.
Observation	Separable Gaussian wavelength/x/y PSF preserves the high-resolution cube shape.	Delta response, normalization and full-grid tests.
Weights	Wavelength-by-Stokes and spatial weights multiply; no separate Stokes-weight field.	Zero-weight exclusion and residual packing tests.
Configuration	Repeatable regions/sources, dataset paths, grid spacing, physics capability and optimizer controls are validated.	43 configuration/checkpoint assertions in the current suite.
Checkpoint	Versioned atmosphere, solver state, capabilities and metadata; topology-independent payload.	Serialization round trip and 1-rank/4-rank restore tests.
Extension seams	Abstract population, redistribution, opacity, formal-solver and observation interfaces.	Mock PRD/IQUV traversal and explicit unsupported-capability errors.

Configuration decisions retained

- No implicit shape relationship is imposed between observation-active pixels and the forward grid.
- A 200 x 200 observed selection can be fitted from an 800 x 800 forward cube by supplying an 800 x 800 spatial-weight map with zeros away from selected pixels; wavelength selection works the same way.
- Only controls with an implemented role remain. Included optimization controls cover iteration count, chi-square stopping, damping, SVD threshold where applicable, and bounded random perturbation; STiC legacy switches without a defined Julia behavior are omitted.
- The initial atmosphere supplies log tau500 directly. HE3D/MHS derives geometrical z; it does not replace the inversion coordinate.

Key records

ADR-001 through ADR-010 capture coordinates, the Python bridge, force balance, future PRD/Stokes seams, observation grids, the two-input/two-output contract, vertical regularization, mixed synthesis, the hybrid runtime, and the bounded Phase 5 prototype solver.

5. Phase 1 - HE3D/MHS and thermodynamics

Mode selection is structural. If B is absent, magnetic arrays and Lorentz work do not exist. If all three B components are present, the solver computes J = curl(B)/mu0 and includes J x B, changing the mode to MHS.

Component	Implementation	Important refinement beyond the plan
Pressure	Vertical integration initializes a 3D Jacobi-like least-squares gradient relaxation over x, y and z neighbors.	This is not a set of independent 1D columns; the boundary plane remains fixed.
EOS	Wittmann C++ backend maps T and Pgas to rho and ne through Julia FFI.	Reference ideal-gas EOS remains only for manufactured tests.
500 nm opacity	STiC cop continuum evaluated at 5000 Angstrom, converted from cm^-1 volume extinction to m2/kg mass opacity.	Reference opacity is explicitly non-production.
Height	Opacity/density integration reconstructs corrugated z while retaining log tau500 as the inversion grid.	Temperature and optional B stay exactly on target optical-depth points.
Convergence	Independent pressure, density, force-residual and height-change thresholds with under-relaxation.	EOS or force-balance failure is an error, not silent clipping.
Distributed form	Halo derivatives, global z-coordinate reduction, global maxima and topology-stable pressure sweeps.	Jacobi updates were chosen to make rank ordering irrelevant.

Validated numerical facts

- Phase 1 manufactured tests: 12 of 12 pass; production Wittmann/STiC tests: 6 of 6 pass.
- At 5770 K, the compiled backend returns positive finite rho and ne for Pgas = 1 Pa and 100 Pa.
- The production 500 nm mass opacity agrees with the independent Python Wittmann path at five points from 3500 to 12000 K and 0.1 to 1000 Pa, with maximum relative difference below 3e-4.
- The MHS fixture has nonzero Lorentz force, horizontal pressure structure, a pressure solution distinct from HE3D, and exact B preservation.

Known scientific discretization limit

Curl uses declared x/y coordinates and a horizontally averaged geometrical-height coordinate. This orthogonal-grid stencil must be replaced or scientifically validated with metric terms before interpreting strongly corrugated MHS volumes.

6. Phase 2 - FFNO population service

The population layer is a persistent, substitutable service. PythonCall is a weak dependency; CPU-only HE3D, recorded inference, configuration, and synthesis tests do not initialize Python.

Contract	As built
Input features	temperature, vx, vy, vz, log10(ne), log10(rho) in canonical (channel,nz,nx,ny) order; corrugated z and physical dx/dy are separate.
Output	Linear populations in m^-3 with canonical (nz,nx,ny,nlevels) order.
Persistence	Model factory, normalization, checkpoint and device state load once and serve repeated in-memory calls.
Validation	Channel order, normalization sizes, level map/count, checkpoint SHA-256, finite inputs, and positive populations are checked.
CPU fixture	RecordedPopulationModel stores an exact feature/z request and response; mock and recorded models share predict_populations!.
Distributed integration	Rank 0 gathers the globally coupled feature volume, launches the service through the GPU coordinator, then scatters population tiles.

Checkpoint evidence

Model	SHA-256	Metadata and parity
H	901dcd28a6ee651c...e87803c755	Epoch 171; validation loss 0.0058831; Cin=6, Cout=6. Bitwise identical to legacy untiled FFNONet after the declared axis permutation.
Ca	7a9365d26543dcb...aa4af9b760	Epoch 252; validation loss 0.00730947; Cin=6, Cout=6. Loads and returns finite positive populations.

Local runtime scope

The supplied H and Ca checkpoints load and run on CPU. Approximate tiny-fixture inference was 7.6 s for H and 7.4 s for Ca, with about 0.9 s H construction/load. These values prove integration and persistence only; they are not GPU performance targets.

Not locally testable

No CUDA/MPS device was available. Device placement, GPU peak memory, host-device transfer overlap, real multi-node rank-0 launch under Slurm, and GPU throughput remain target-hardware evidence.

7. Phase 3 - mixed non-LTE and Kurucz LTE synthesis

The implementation treats a spectral region as an ordered list of opacity/emissivity contributors, not as one exclusive LTE or non-LTE mode. Halpha and Ca II 8542 may consume FFNO populations while Kurucz lines such as Fe I 6301/6302 are synthesized in LTE during the same forward evaluation.

Layer	Implementation decision
Line source	FFNOTransition and KuruczLTEModel implement a common add_opacity_emissivity! seam.
Combination	Contributors add extinction and emissivity on one high-resolution wavelength/depth grid before one formal solve.
Continuum	Exactly one contributor owns continuum opacity; remaining contributors add line terms, preventing double counting.
Kurucz parsing	Portable K94 fixed-width parsing includes species/stage, level ordering, statistical weights, Einstein coefficients and tabulated damping.
Wavelength	Configured STiC-style observation wavelengths remain air wavelengths; line physics uses RH air-to-vacuum conversion internally.
LTE state	Persistent Wittmann partition/Saha state computes ionization, excitation, neutral hydrogen and wavelength-dependent continuum.
Profile	Armstrong Voigt and RH RLKProfile velocity, Doppler and damping terms are ported; lists are parsed and window-selected outside the hot loop.
Formal solution	Production Muspel adapter plus a scalar reference solver; output is always SpectralCube with a Stokes axis.

Evidence

- Phase 3 mixed-synthesis tests: 15 of 15 pass inside the current full suite.
- Real K94 parsing and the supplied kurucz_6301_6302.input file are exercised, including wavelength convention, species decoding, window selection, positive finite spectra, blends, and LTE-only setup without FFNO.
- Using the supplied snapshot 385 reference data and Tiago H/Ca atom models, Halpha and Ca II 8542 at column (1,1) are bitwise identical to stored intensities: maximum absolute and relative error are both 0.0.
- PRD and non-I synthesis are explicitly rejected by the current mixed backend rather than silently approximated.

External-reference boundary

No independent stored Kurucz reference spectrum was supplied. Kurucz acceptance is therefore source-level parity with the RH equations plus real-line-list and manufactured-atmosphere tests, not parity with an independent 6301/6302 golden cube. Zeeman, isotopic/hyperfine structure and RH scattering remain outside this release.

8. Phase 4 - hybrid MPI, threads, and rank-0 GPU

Phase 4 integrates the Phase 0-3 components into one distributed application path. MPI calls are confined to the thread that initialized MPI. Julia threads use static column scheduling and thread-private mutable workspaces; BLAS/OpenMP thread policy remains a launch responsibility.

Subsystem	Distributed behavior	Why this decision
Decomposition	Near-square, non-empty 2D Cartesian x/y tiles; uneven sizes are supported deterministically.	Reduces surface/volume ratio versus a fixed slab and supports x/y spatial operators.
Transport	Typed MPI buffers for atmospheric/spectral payloads; object serialization only for small control metadata.	Avoids language serialization in the scientific data path.
Halos	Exchange x first, then y including new x halos; physical edges replicate boundary values.	Produces correct corners for MHS, PSF and horizontal regularization.
Threads	Each rank processes its tile columns with Threads.@threads :static and thread-private extinction/emissivity workspaces.	Allows, for example, 16 ranks x 16 threads on 256 CPUs without 256 MPI processes.
FFNO control	Rank 0 opens a TCP status channel, then alone launches the nested/persistent GPU operation; MPI ranks wait outside a long collective.	Adapts the successful charge-branch pattern and avoids nested Slurm/NCCL conflicts.
Failure	Rank 0 broadcasts success/error through TCP; all ranks pass a recovery barrier and throw the same failure.	Prevents a launcher exception from stranding other ranks.
Objective	Chi-square and regularization scalars use reductions; horizontal terms count uniquely owned forward differences.	Retains tile-local residual/atmosphere arrays while producing a global objective.
Outputs	Final spectrum and atmosphere currently gather to rank 0.	Correct root adapter now; parallel hyperslab writing is a later I/O optimization.

Instrumentation added for acceptance

- HybridForwardTimings records force-balance, population, synthesis, observation and total wall time.
- distributed_memory_report records per-rank tile shape and owned atmosphere/workspace bytes for the hot path.
- parallel_provenance records Julia version, MPI/thread topology, process grid, launcher rank, host/rank/tile map, capability list, and supplied configuration/model/source hashes; it can be written as stable TOML.

9. Phase 4 local verification evidence

All results below were produced on 25 Aug 2026 from the working tree represented by this report. The MPI fixture includes B, a nonzero Lorentz term, 3D force balance, a globally coupled population model, threaded synthesis, spectral and spatial PSF, observation distribution, chi-square, and vertical/horizontal regularization.

Evidence item	Test design	Observed result
Complete package suite	julia --project=. --threads=4 test/runtests.jl	180 of 180 assertions passed across 14 test sets.
Full-field topology parity	One rank x two threads versus four ranks x two threads on nz=4, nx=11, ny=7.	10 full fields agree: spectrum, populations, T, Pgas, rho, ne, z, Bx, By, Bz. rtol=1e-12, atol=1e-14; regularization rtol=1e-13.
Global FFNO staging	Population fixture depends on the mean temperature of the complete global atmosphere.	Same population cube and final spectrum across one and four ranks; checksum 5.458701579229605e12.
Memory ownership	Before final gathers, assert unique tile coverage, tile-local populations, and no full model on any rank for N>1.	All assertions pass. Largest four-rank fixture hot-path owned array total: 15,744 bytes.
Allocation stability	Warm repeated forward! calls; compare workspace object identities and allocated bytes.	3,637,520 bytes on first measured call and 3,637,520 on second; all identities stable; spectra exactly equal.
GPU-control failure	Rank-0 launcher throws intentionally; all four ranks catch; second launch returns 42.	MPI_PHASE4_FAILURE_RECOVERY_OK ranks=4.
Restart topology	Write 1 rank -> read 4; write 4 -> read 1; compare full atmosphere fields.	Both directions pass exactly.

Topology parity output

```
1 rank: checksum=5.458701579229605e12, regularization=4.0000000000000544e-4
4 ranks: checksum=5.458701579229605e12, regularization=4.000000000000052e-4
PHASE4_FULL_FIELD_PARITY_OK fields=10 nx=11 ny=7 max_owned_bytes=15744
```

Interpretation

The tiny regularization difference is floating-point reduction ordering and is below the declared tolerance. Spectrum and globally coupled populations match. The uneven grid exercises unequal tiles and corner halos, not only a divisible demonstration case.

10. Local performance and memory observations

A same-core-count smoke benchmark compared three launch layouts. Each layout used the same 11 x 7 scientific fixture and produced the same spectrum checksum. Cold wall time includes Julia specialization and process startup; instrumented forward time begins inside forward!.

Ranks x threads	Cold wall (s)	Forward (s)	HE3D/MHS (s)	Populations (s)	Synthesis (s)	Observation (s)	Max owned bytes
1 x 4	4.505	0.695	0.652	0.034	0.0020	0.0051	47,728
2 x 2	4.883	1.018	0.762	0.235	0.0019	0.0167	25,824
4 x 1	5.054	1.049	0.777	0.256	0.0018	0.0117	14,016

Local conclusion

For this very small single-node fixture, 1 rank x 4 threads is fastest because MPI setup, root staging and halos dominate the available work. Increasing rank count reduces per-rank owned arrays monotonically: about 47.7 kB -> 25.8 kB -> 14.0 kB. This validates the expected memory distribution but is too small to recommend the production topology.

Allocation interpretation

Workspace identity reuse is proven, but the forward call is not allocation-free: 3.638 MB is still allocated per measured evaluation in the one-rank fixture. The remaining allocations arise from iterative force-balance and halo/observation temporaries. They are a performance optimization opportunity, not a correctness blocker; production profiling should decide which buffers merit promotion into reusable workspaces.

Topology rule for the first cluster run

- Keep total Julia threads per node within allocated physical cores and set external BLAS/OpenMP thread counts deliberately to avoid hidden oversubscription.
- Start with fewer MPI ranks per node and several Julia threads per rank, because the architecture was chosen to reduce inter-rank communication while still using all CPUs.
- Measure at least 1, 2, 4, 8 and 16 ranks per node with fixed total cores, then weak/strong scaling across nodes. Report stage time, communication/control time, peak host/device memory and efficiency.
- Choose the production layout from the real grid, line list, wavelength count and FFNO checkpoint; do not transfer the tiny-fixture 1 x 4 result directly.

11. Engineering decisions made during implementation

These refinements are consistent with the development plan but were decided while converting it to working code.

Decision	Reason and consequence
Near-square 2D tiling	The process grid minimizes tile aspect distortion while requiring non-empty tiles. This supports both x and y operators and makes ownership deterministic for uneven domains.
Main-thread-only MPI	The package does not require MPI_THREAD_MULTIPLE. Threaded kernels complete before communication; accidental MPI from a worker thread fails immediately.
Static threaded scheduling	Column work uses predictable thread ownership and private scratch arrays. Cache size uses available Julia thread slots rather than a shared global buffer.
Jacobi-style distributed pressure sweeps	Each sweep reads one halo-consistent pressure state and writes the next. This removes rank update-order dependence and enabled strict one/four-rank parity.
Corner-complete two-stage halos	Y exchange includes x halos, transferring corners without separate diagonal messages. Physical edges use replicated values consistently in serial and MPI modes.
TCP GPU status plane	The nested GPU launcher does not hold outer MPI collectives open. Host may be overridden for cluster routing; connect retries cover startup skew.
Globally coupled population fixture	The Phase 4 MPI test uses a population function of the global temperature mean. A pointwise mock would not prove rank-0 full-context staging.
Root final-output adapter	Collecting final products is correct and simple now. Collective HDF5 can replace only the adapter, without altering physics or rank ownership.
TOML provenance	Small rank/tile metadata may be gathered as Julia objects; scientific arrays always use typed MPI buffers. Hash values are accepted from the caller so run management controls their definition.
Mixed contributors, one solver	Kurucz LTE is not a parallel synthesis program. LTE and FFNO sources join before one formal solution and one PSF/objective path.

12. Phase 5 - bounded small-volume inversion

Phase 5 closes the loop around the Phase 4 distributed forward path. Rank 0 owns optimizer decisions and atomic checkpoints; only the small coarse control vector is replicated. Every atmosphere, population, spectrum and residual remains rank-tiled while objective components are globally reduced.

Component	As-built behavior	Evidence
Control maps	Temperature, vx/vy/vz or vlos, microturbulence, and optional Cartesian B controls use vertical nodes plus coarse x/y maps. Every block has physical bounds and a positive optimizer scale.	Packing, projection, expansion and coarse-to-fine interpolation tests pass.
Distributed objective	A trial restores its thermodynamic reference, expands controls directly onto the local tile, executes the hybrid forward path, and reports normalized weighted chi-square and regularization separately.	No full atmospheric cube is introduced into the solver.
Prototype optimizer	Deterministic best-improvement coordinate pattern search in scaled coordinates; rank 0 proposes/accepts and all ranks evaluate the same trial.	Two starting temperatures converge to the same six-parameter truth.
Gradient oracle	Centered directional finite differences operate in normalized coordinates and reject bound-clipped stencils.	Steps 1e-3 and 5e-4 agree within 1e-8 relative tolerance.
Checkpoint	Rank 0 atomically replaces a capability- and layout-validated serialized state after completed iterations.	Interrupted 3 -> 8 iteration history equals uninterrupted 8 iterations.
Diagnostics	Rank 0 writes summary.toml and objective_history.csv with final controls, topology, objective components, accepted coordinate/direction and evaluation count.	File content and history-row count are asserted.

Synthetic recovery gate

The exact-model fixture uses a small 3D intensity cube with independent responses to upper/lower temperature and line-of-sight velocity. Six controls encode a two-point horizontal temperature mode and a two-node vertical velocity mode. From 4500 K and 7000 K initial temperatures with zero velocity, the objective falls by more than six orders of magnitude and every control reaches the known truth exactly on the configured 500-unit search lattice.

MPI and restart evidence

One rank and four ranks execute eight optimizer iterations with identical parameters and accepted decisions. Objectives differ only by 4e-15 from reduction ordering. A checkpoint written after three iterations with one rank resumes with four ranks to the same final parameters, history and evaluation count as the uninterrupted run.

Prototype boundary

Pattern search is intentionally a small/coarse reference method. This exact-model result validates orchestration and fixture identifiability; it does not establish recovery with FFNO/Muspel/Kurucz model mismatch, noisy observations, GPU gradients, or production-size fields. Phase 6 supplies matrix-free VJPs and a scalable solver; Phase 7 supplies independent RH/MULTI3D evidence.

13. Phase 6 - production gradients and scalable solver

Phase 6 replaces the prototype search with a production objective pullback and bounded L-BFGS while retaining the sole MPI/thread/GPU forward architecture. Gradients use scaled coarse controls and never allocate a dense spectral Jacobian.

Component	As-built derivative	Evidence
Observation	Exact transpose of weighted chi-square and separable spectral/x/y Gaussian degradation, including MPI boundary exchange.	Directional dot product passes.
Transfer and lines	Exact scalar formal-solver reverse; analytic FFNO-transition and Julia Kurucz LTE pullbacks; cell-local Muspel and opaque Wittmann fallbacks.	Mixed two-species non-LTE plus Kurucz objective agrees with centered finite differences to 3.67e-10 relative error.
FFNO	Persistent PyTorch VJP exposed through PythonCall. Population cotangents gather to rank 0; feature/z cotangents scatter back to owning tiles.	One-rank and four-rank exact MPI VJP fixture passes; non-root ranks own no model.
Force plus regularization	Bounded centered action on the CPU reconstruction/regularization composite only; one complete forward/GPU call per objective gradient.	Complete Taylor remainder is second order.
Solver	Projected bounded L-BFGS, Armijo trials, limited-memory checkpoint/restart, safe restoration after all rejected trials.	Local, MPI, cross-topology restart and Olivia recovery probes pass.
Executable contract	HDF5 axes time/z/y/x and time/Stokes/wavelength/y/x feed one MPI runner and exactly two configured scientific outputs.	Seven end-to-end HDF5 assertions pass.

Local and MPI gate

The full suite passes 220 assertions. The mixed production gradient reduces its objective with L-BFGS in 22 complete forward evaluations. The MPI validator passes for one and four ranks and restarts a one-rank checkpoint on four ranks with matching controls, history and evaluation count.

Olivia gate

Job 2072078 passed distributed L-BFGS, rejected-trial recovery, the 800 x 800 rank-owned memory probe, and CUDA autograd/NCCL VJP. Maximum reported step RSS was 10.7 GiB. The updated harness adds a real H-checkpoint VJP versus finite-difference test; this post-implementation regression still needs one Olivia rerun.

14. Evidence that cannot be produced locally

These items need hardware, scheduler access, or an independent reference artifact unavailable on this workstation. They are not represented as passing tests.

Missing evidence	Why local execution is insufficient	Required target test / acceptance
Post-change real FFNO VJP	The workstation has no CUDA/PyTorch runtime for the production checkpoint.	Run the updated Phase 6 Olivia group and retain OLIVIA_PRODUCTION_FFNO_VJP_OK plus its diagnostic archive.
Production topology recommendation	The 11 x 7 test is latency dominated and much smaller than the intended atmosphere/spectral cube.	Benchmark real dimensions and line mix at fixed cores per node; report strong/weak scaling and choose ranks x threads empirically.
Parallel HDF5 output	Current output adapter gathers to rank 0; no collective HDF5 backend was needed for local correctness.	Implement only if output collection exceeds memory/latency limits; test hyperslab ownership and bitwise reconstruction.
Independent Kurucz golden spectrum	The supplied file is a line list, not a trusted external spectrum for the same atmosphere and conventions.	Produce RH/STiC 6301/6302 spectrum and compare isolated lines, blends, continuum and wings with declared tolerances.
Strongly corrugated MHS metric	Current curl is an orthogonal-grid approximation using horizontally averaged z.	Manufactured curvilinear solutions and an independent MHS reference must validate or replace the operator before scientific use.

Deployment status

Phase 6 source implementation and local/MPI evidence are complete. Olivia job 2072078 accepted the existing target runtime; the new real-checkpoint VJP case is a required post-change regression, not a new architecture task. Independent-physics accuracy belongs to Phase 7.

15. Reproduction commands and evidence map

Run from /Users/harshm/Documents/WorkRepo/FNOML/FFNOInversion.jl. The libxml version warning emitted by local MPI is recorded but did not cause a test failure.

```
# Complete threaded suite
julia --project=. --threads=4 test/runtests.jl

# Phase 6 gradient/solver topology and cross-topology restart
julia --project=. scripts/validate_phase6_mpi.jl

# Execute the two-input/two-output inversion
julia --project=. scripts/invert.jl CONFIG.toml MODEL_FACTORY.jl

# Submit bounded target-runtime and real-checkpoint VJP tests
bash scripts/submit_olivia_phase6_tests.sh

# Halpha and Ca II 8542 stored reference parity
julia --project=. scripts/validate_muspel_reference.jl
```

Implementation map

Concern	Primary source / evidence
Parallel context, tiles, scatter/gather, halos, GPU control, provenance	src/Parallel.jl; test/parallel_runtime.jl; test/mpi_hybrid_worker.jl
Distributed force balance, FFNO staging, synthesis, PSF, objective, memory	src/DistributedForward.jl; test/phase4_integrated.jl; test/mpi_phase4_worker.jl
HE3D/MHS, EOS and opacity	src/ForceBalance.jl, EOS.jl, Opacity500.jl; phase-1-validation.md
Population bridge	src/PopulationModels.jl; ext/FFNOInversionPythonCallExt.jl; phase-2-validation.md
Mixed line synthesis	src/LinePhysics.jl, Synthesis.jl; ext/FFNOInversionMuspelExt.jl; phase-3-validation.md
Config, checkpoints and scientific contracts	src/IO.jl; test/io_config.jl; ADR-005 through ADR-008
Phase 4 validators	scripts/validate_phase4_topology.jl, validate_phase4_restart.jl, benchmark_phase4_local.jl
Phase 5 controls, objective and solver	src/Solvers.jl; test/phase5_inversion.jl; ADR-010; phase-5-validation.md
Phase 5 MPI/restart validator	scripts/validate_phase5_mpi.jl; test/mpi_phase5_worker.jl
Phase 6 production pullbacks and solver	src/ProductionGradients.jl, Gradients.jl, ScalableSolvers.jl; test/phase6_production_gradient.jl
Phase 6 HDF5 execution	src/Execution.jl; scripts/invert.jl; test/phase6_execution.jl
Phase 6 MPI/GPU evidence	scripts/validate_phase6_mpi.jl; test/mpi_phase6_worker.jl; run_olivia_runtime_tests.sbatch

16. Handoff to Phase 7

Phase 6 implementation is complete. After retaining the updated Olivia real-checkpoint VJP marker, the active development phase can move to independent-physics validation.

Phase 7 should begin with

- Generate held-out spectra and populations with MULTI3D/RH rather than the FFNO inversion surrogate.
- Compare populations, spectra and recovered atmospheric variables by height, wavelength and spatial scale.
- Separate FFNO population error, force-balance mismatch, Kurucz/Muspel synthesis mismatch and optimizer error.
- Agree quantitative scientific acceptance thresholds and trigger the documented retraining contingency if they fail.

Stable boundaries entering Phase 7

Stable boundary	Reason
One MPI/thread/GPU architecture	The optimizer must call the existing distributed forward path; no serial-only solver branch.
Rank-owned atmospheric state	Optimizer parameters and gradients remain distributed; rank 0 owns only global scalars, control and permitted staging.
Two-input/two-output contract	Solver state and diagnostics remain sidecars, not new primary scientific files.
Full-grid synthesis plus weights	The optimizer never resizes/interpolates to choose data points; PSF precedes weighted chi-square.
Optional B	No-B inversions remain HE3D and do not allocate magnetic parameters. B-present runs select MHS and later Stokes capability independently.
Intensity/non-PRD capability seams	Current I/non-PRD physics remains active while the interfaces preserve future PRD and IQUV.

Acceptance statement

Phase 6 source implementation is complete and its local, MPI and prior Olivia infrastructure gates pass. Final sign-off requires one rerun of the updated Olivia group to retain the production H-checkpoint VJP marker. Once received, Phase 7 can start without an architectural change.

Companion documents

Planning baseline: output/pdf/ffnoml_julia_3d_inversion_development_plan.pdf
Implementation source: FFNOInversion.jl/
Phase reports: FFNOInversion.jl/docs/reports/
Architecture decisions: FFNOInversion.jl/docs/adr/